Chapter 6

Teaching AI ethics for translation students

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A key part of teaching translation in the age of Generative AI (GenAI) is understanding when to use and when not to use GenAI. AI ethics is an applied subfield of Ethics that can help with decision-making, which may be useful to students after they graduate and as they move through their careers. In this chapter, we begin with some definitions of AI and ethics, introduce some ethical issues, along with some progress in addressing those issues. We finally consider some ways to introduce discussion and reflection in the classroom to maximise the impact of this teaching, focusing in particular on case studies and the use of mapping and images to understand our position within the complex interrelationships between individuals, organisations, and society regarding ethical issues pertaining to translation and GenAI.

1 Introduction

The topic of artificial intelligence (AI) ethics is very high on many agendas at present, as universities and schools grapple with how best to incorporate Generative AI (GenAI) into their curricula in a constructive manner. The intention is to teach AI literacy, defined by Long & Magerko (2020: 598) as the ability to critically evaluate, communicate and collaborate with, and use AI, equipping students with critical knowledge and skills that will stand to them throughout their careers, balanced with understanding of ethical issues regarding the use of AI.

AI ethics are important. There are lots of ethical issues regarding GenAI, as we will run through in this chapter. Our graduates will gain responsibilities as they

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continue through their careers, so it's important that they be conscious of the ethical repercussions of decisions and actions that they might take. The current hype about GenAI might mean pressure to use tools even if the circumstances are not appropriate – what a blogpost by the language service provision company Transperfect called the "push to implement", when companies are "pressured by leadership to implement AI solutions" (Transperfect 2024). It's important that graduates are able to make well-founded arguments as to why using GenAI might be appropriate or inappropriate in any given situation.

In the classroom, questions of ethics tend to provoke discussion and allow for student input. This chimes with the call for critical and democratised education from authors such as Freire, or Giroux and McLaren, incorporating "forms of learning which serve to prepare students for responsible roles as transformative intellectuals, as community members, and as critically active citizens" (Giroux & McLaren 1997: 236). Currently, this call feels more urgent than ever. In Translation Studies, Abdallah (2011) presented a related three-step 'ideology critique model for teaching' that involves disassembly of the teachers' own beliefs, critiquing and resisting unfairness, and finally fostering hope and encouraging agency. The aim of teaching about ethics in the translation classroom is to stimulate critical thinking so that students recognise ethical problems and can resist unfairness. However, consideration of ethics may present students with opportunities rather than only imposing limitations, and it is important for students to graduate with a positive sense of their own agency in ameliorating ethical issues. Equally, as educators and students, we should be aware of and take responsibility for our choices and decisions. With that in mind, this chapter borrows Bryan's (2022) 'pedagogy of the implicated' to describe and understand our position within the complex interrelated systems linking us as individuals to our surrounding political, cultural, social and economic systems.

This chapter will begin with some definitions of terms pertaining to AI and ethics, moving on to explaining some of the ethical issues. We have found it useful to create model use cases for in-class discussion and will suggest a couple of these. Thereafter we look at possible routes forward regarding GenAI and ethics and the recent progression on these. Finally, we look at ethical issues for researchers in translation and interpreting that might be relevant for both educators and students.

2 Definitions

The terms 'artificial' and 'machine intelligence' were first popularised in the 1950s, usually regarding the ability of machines to take on human tasks, but are rarely clearly defined. More recent definitions of AI are still sometimes related to thinking or behaving like humans or thinking or behaving in a rational way. Russell (1999: 12), for example, defines AI as "the capacity to generate maximally successful behaviour given the available information and computational resources". Up until the 1980s, these resources might have involved sets of rules built by human experts – so-called 'expert systems'. This was the common paradigm of rule-based machine translation (MT), before the advent of statistical MT (Brown et al. 1988).

Both rule-based and statistical MT used human-readable data or rules in what is sometimes called Symbolic AI, but once technologies such as MT, speech recognition, and recommendation systems moved to neural networks or machine learning, we moved to Subsymbolic AI, in which data and processes are transformed into numbers and no longer comprehensible by humans. Mitchell (2020: 12) defines subsymbolic AI as a "stack of equations – a thicket of often hard-to-interpret operations on numbers".

In recent times, AI is often taken to mean machine learning using transformer-based neural networks, despite the term's relatively long history. Machine learning is defined by Kelleher et al. (2015: 3) as "an automated process that extracts patterns from data", and can be supervised, whereby the system learns how to carry out a repeated task based on training data, as with NMT, or self-supervised (i.e. by the system itself), where patterns are inferred from training data without explicit external instruction, as is the case with large language models (LLMs) – GenAI based on natural language processing.

Currently, much of the focus for very large (sometimes called 'frontier') LLMs is scale: bigger language models (LMs) with more data, as a key finding for LLMs was the emergence of unexpected abilities at scale (Wei et al. 2022). This does not mean that smaller LMs cannot be useful (see Moorkens et al. 2025 for more on local, small LMs), but the tendency towards larger LMs brings particular ethical concerns that we will return to in the following section.

So, what do we mean by ethics? Ethics is a branch of philosophy that helps us to decide if an action is right or wrong (see Moorkens 2024). Normative ethics describes theories to help us with that decision, such as consequentialism or deontology, and applied ethics involves the application of those theories to particular scenarios, for example in business ethics, data ethics, or AI ethics. The field of AI ethics looks to balance the benefits of AI with negative effects (which we will come to in Section 3), often by establishing principles for providers to adhere to at their discretion or by establishing governance rules within an organisation or in national (or international) law.

There's a growing sense that technology affects our ability to live a good life. Technologies have affordances, actions that are made available to us as users, and these can encourage positive or negative, healthy or unhealthy behaviours. Machine learning is enabled by powerful computers, built using rare metals and electrically powered, and also by huge amounts of proprietary and/or public data. The terms 'crawling', 'harvesting' or 'scraping' are often used for the collection of data via the internet, as if data is naturally-occurring rather than the product of human effort. So there are lots of ethical issues pertaining to AI and data that relate to translation. We'll discuss many of these in Section 3.

Discussion point: How might translation technologies or GenAI help us to live a better life? And might they also have negative effects? If so, what might these be?

3 Ethical issues pertaining to GenAI and translation

As noted previously, the availability of vast amounts of data facilitates GenAI. Translators who work with computer-aided translation tools are usually expected to include bilingual translation memories with delivery of translation jobs, either due to contracts or precedent. This draws from the Berne Convention, first enacted in 1886, in which translations are considered to be derivative works that "shall be protected as original works without prejudice to the copyright in the original work" (World Intellectual Property Organisation 1979: art. 2). There have been arguments that translators may have reasonable claims for copyright (see Moorkens & Lewis 2020), but this is not usually respected, and in many cases, digital platforms do not permit translators to even access these memories, nor to control how they are repurposed. MT and LLMs are trained using monolingual data from organisations' repositories and data 'harvested' from the internet using crawlers. The legal basis for this differs in different jurisdictions and many copyright owners for text crawled or harvested have taken legal action. For now, there does not appear to be any major restriction to using crawled data, even if it's copyrighted, perhaps because authorities do not want to be seen to stifle innovation and few content owners feel able to challenge powerful technology companies. For translators and writers whose data is used, this presents a problem in that their work may be put towards purposes that they do not agree with, without any further compensation.

Workers in translation and other professions may choose to use GenAI or MT (see Rivas Ginel & Moorkens 2024), but many have these technologies imposed on them without any choice. Fırat (2024) and others demonstrate the relationship between the use of technology to reduce labour costs, particularly within digital platforms, and reduced access to what the International Labor Organisation term 'decent work', relating to pay, representation, time, stability, equality and management (Ferrero et al. 2015). Ruokonen & Svahn (2024) summarise numerous studies that show a relationship between increased technologisation and reduced job satisfaction and motivation, which threaten the sustainability of the translation industry. Beyond translation, teams of (ghost-)workers, little-documented and often with poor pay and conditions, work to mitigate biased and offensive output from GenAI as part of the process of reinforcement learning using human feedback (RLHF) or to moderate potentially offensive training data (Rowe 2023), with associated repercussions on workers' mental health.

Some years after NMT became popular, researchers such as Vanmassenhove et al. (2018) found patterns of bias in NMT outputs. This output tended to be less lexically diverse than human translations and contained gender bias, with certain words tending to be associated with males or females. There is also likely to be a cultural or linguistic bias towards English, the language of most training data (Bender 2011, Moorkens et al. 2025). The uneven support for languages in the digital domain is likely to exacerbate existing digital divides. Researchers working with LLMs also found biased output regarding race, gender, and sexuality, and a tendency to overrepresent hegemonic views. This makes sense, considering that a lot of training data comes from the internet. In order to rebalance this and to prevent embarrassing culturally inappropriate or offensive output, developers of LLMs began to use RLHF (Ouyang et al. 2022). However, this intervention also offers LLM developers the opportunity to suppress output due to geopolitical concerns. According to the New York Times, the output of the Chinese GenAI tool Deepseek "will largely reflect the worldview of the Chinese Communist Party" (Myers 2025: B1) and US-based GenAI Grok censored "unflattering facts about President Donald Trump" and its owner, Elon Musk on its initial launch (Wiggers 2025), one of a string of controversies that appears to relate to adjustments of the RLHF guardrails behind the scenes. Cooperation between big tech companies and autocratic regimes has until now mostly involved removing content or search results (see Boyle 2025), but RLHF offers another opportunity to intervene and manipulate output based on political demands.

Translation and MT have long been closely related to power (see Tymoczko 2007 and Paullada 2020) with the patron of translation or MT research having some sway in production. The amount of data, money, and computing power required to train LLMs mean that production capacity is in the hands of very few wealthy organisations, with many produced by US big tech companies or their close relations. Users are further reliant on these companies' cloud services to train and fine-tune, and later deploy their LLMs. Services may be changed or withdrawn with little warning due to a change of circumstances (or geopolitical expediencies). At the time of writing, there is a push for LLM sovereignty, with international organizations like the European Union, national governments and some companies investing in supercomputers and building their own LLMs. Of course, not all LMs have to be huge, but the focus for US big tech providers continues to be scale, creating larger and larger models that take longer to train, with knock-on effects regarding environmental sustainability.

The pretraining and deployment of foundational LLMs is a resource-intensive process that requires significant capital investment for acquiring the necessary hardware, training data and human talent as well as operating costs, and each large-scale pre-training cycle has significant environmental impacts due to high electricity and water consumption. Combined, these factors allow only a few companies to have very powerful LLMs. Since many national governments, large corporations and public organizations consider LLMs and AI in general as strategic enablers for their future, they are increasing their investment in this area as well and setting big targets for broader AI use. However, the impact of this AI 'arms race' on the environment is likely to be catastrophic on our planet if every entity tries to develop their own proprietary LLM. In recent years, AI systems have in general become more efficient, but the trend for scale means quality improvement is based on more data and parameters, requiring more training time. Luccioni et al. (2025) argue that increased efficiency in AI training and in time saved by using AI will tend to spur more use of AI, continuing an upward trend not only in emissions, but in water use for data centres, rare earth metals used for ICT, and more harmful electronic waste.

These environmental harms and the impacts directly from AI are difficult to measure exactly. Emission levels may differ depending on whether energy comes from fossil fuel or renewable sources (Shterionov & Vanmassenhove 2022) or depending on the time of day and data centre location (Dodge et al. 2022). Big tech firms are looking to buy up renewables and nuclear power sources as they come online, in order to maximise their net-zero credentials. Water footprints are also likely to differ depending on time and region (Li et al. 2025). Presently, GenAI does not require a large proportion of resources, but projections, such as those from the head of the UK National Grid predicting an AI-driven six-fold increase in power requirements for data centres in the next decade (BBC 2024), are worrying. For now, energy and water use attributable to GenAI are very small in comparison to the huge requirements for watching streaming media or joining a Zoom meeting (Mytton et al. 2024).

4 Possible solutions

Positioned within the international AI ‘arms race’, with politically-motivated support for some developers leading to preferred companies receiving preferential access to governments and cutting-edge technologies, there are also movements to use ‘AI for Good’. One platform by that name is led by the International Telecommunication Union of the United Nations (UN) to use AI to achieve UN sustainable development goals, and there are smaller initiatives such as the Distributed Artificial Intelligence Research Institute that looks to push back against the influence of big tech on AI research, development and deployment. Many people seek to use (Gen) AI to reduce harms and inequality. There have been proposed uses of AI to route power use to maximise the use of renewable energy and to improve energy efficiency in the design, building, and use of commercial buildings (Ding et al. 2024).

Van Wynsberghe (2021) feels that there are two motivations behind ‘AI for sustainability’ and ‘sustainability of AI’ that ought to be combined. The former seeks to do good, yet might entail negative environmental impacts, whereas the latter acknowledges that for AI to be sustainable, there needs to be lower environmental costs for AI training, tuning and inference. She defines sustainable AI as a necessary movement to “foster change in the entire lifecycle of AI products (i.e. idea generation, training, re-tuning, implementation, governance) towards greater ecological integrity and social justice” (Van Wynsberghe 2021: 217).

National and international legislation, most notably the EU AI Act, seek to limit harmful uses of AI. The AI Act defines a typology of tiered uses of AI based on risk, with ‘unacceptable risk’ uses forbidden and high-risk uses, such as recruitment decision-making and job allocation, subject to special regulation. This renders algorithmic job allocation in the EU illegal, although it’s likely that the recommendation of an automated project management tool will still be followed.

In a position paper by Moorkens et al. (2024) we borrowed the idea of a triple bottom line from Business Ethics (Elkington 1997) to propose that translation technologies and LLMs be evaluated not just focusing on performance, but rather giving equal weight to people, planet and performance. This is necessarily a heuristic rather than an exact metric, but follows on from criticisms of a focus purely on performance pushing AI development in the wrong direction by Schwartz et al. (2020) and others. For people, we might consider how the LLM impacts annotators, translators, platform workers, and those who have been dispossessed of their data, balancing these against benefits to people. For the planet, we might look at energy costs/CO_2, efficient models, and ICT cost and disposal. Finally, for performance, we should use task-appropriate and comprehensive standards.

A previous suggestion for translation data from Moorkens & Lewis (2019) was a community-owned and managed digital commons, following the ideas of Ostrom (2011), with tiered access available for a cost. We argued that this would help to "sustain the occupation of translation and to minimise the potential risks and harms to translators and the public" (Moorkens & Lewis 2019: 17). This idea seems to be similar to the intended implementation of a European Language Data Space as a decentralised marketplace for text, video and audio data in different languages (Rehm et al. 2024).

The advice from some researchers, such as Rudin (2019), is to entirely avoid black-box, subsymbolic systems for high-risks uses. This is because opacity is largely baked into subsymbolic AI systems, as described previously, although Rudin and others believe that, in many cases, comparable results may be achieved with more transparent, hybrid systems.

For very large, closed source systems, we do not know what training data has been used or what the RLHF guardrails are. However, not all LLMs are closed source – or even that large. We mentioned the alternative options for small-scale LMs in Section 2. These can be useful for narrowly-defined tasks, with benefits of low cost, low environmental impact, and customisability. For technically confident students, guidelines for building a custom small LM are provided by Moorkens et al. (2025).

In academic research ethics, the credo has moved on from 'do no harm' to the need to actually benefit research participants. Relatedly, best practice for engaged research involves a participatory approach, working cooperatively, particularly with marginalised groups, as co-creators of knowledge rather than imposing narratives or putting words into their mouths. Birhane et al. (2022) propose this approach for building AI systems, offering examples of participatory approaches that aim to lessen existing imbalances of power. In this way, developers "acknowledge that the communities and publics beyond technical designers have knowledge, expertise and interests that are essential to the development of AI that aims to strengthen justice and prosperity" (Birhane et al. 2022: 7).

5 Translation and AI ethics in the classroom

5.1 Classroom discussion activities

As student users of translation technology – and most likely users of related AI tools and cloud services more broadly – students in a translation classroom are already part of the interconnected web of ethical issues from the previous section. They may not have given them much thought, as so much of the hype about GenAI focuses on its positive potential rather than ethical issues. GenAI is also entangled with what Brand & Wissen (2021) call the ‘imperial mode of living’, through which public and private organisational strategies and individual lifestyles and practices in the Global North rely on the unlimited appropriation of resources, a disproportionate claim to global and local ecosystems, and cheap labour, ideally from distant locations. Bryan (2022: 330) conceptualises individual relations with the climate crisis as a “form of ‘difficult knowledge’, particularly as it relates to learners’ self-implication in the conditions that are being addressed”. We can reasonably broaden this to AI ethics for trainee translators and educators. Transmission-based lectures alone seem inappropriate for this topic, as difficult knowledge may raise sensitivities; nobody likes to be hectored or to feel that their personal ethics are in question.

In this section, we introduce two methods for stimulating discussion and reflection about translation and AI ethics in the classroom. The first uses scenarios or case studies, placing ethical dilemmas into familiar contexts for discussion. The second draws from Bryan’s (2022) ‘pedagogy of the implicated’, which seeks to prompt critical reflection about our own positioning as ‘implicated subjects’ and to foster agency for change. Bryan uses the notion of the implicated subject from Rothberg (2019) to look beyond dichotomies of individual versus institutional responsibility for injustices to a discussion of how we are enmeshed with systems in many ways across historical (diachronic) and contemporary social-structure (synchronic) lines.

5.2 Case studies

According to Benbunan-Fich (1998), a combination of lectures and discussion are complementary ways of introducing ethical issues in the classroom. To begin with, lectures about “ethical concepts can lay the theoretical foundations” so that students can “practice ethical analyses” thereafter using case studies (Benbunan-Fich 1998: 20). Case studies have proved to be a useful tool, particularly in business schools, for many years. According to Barnes et al. (1994), case studies extend learning beyond each class, stimulating deeper insights that link across classes and modules. Led by instructors with appropriate case studies, students will engage and can develop and articulate critical insights. These come through four particular factors: situational analysis, active student involvement, a non-traditional instructor role and the need to relate analysis and action. Situational analysis means that ethical issues are applied in situ rather than in theory. Students are active, engaged and learning without direct teaching, energised and challenged by discussions. This puts the instructor in a non-traditional role of facilitator without dominating the classroom. Rather than only building knowledge, case studies combine analysis and action: the "importance of action influences the entire case discussion, which focuses on the practical and doable" (Barnes et al. 1994: 48).

The case study chosen must be appropriate – Zhou (2022: 400), for example, recommends finding "representative translation-ethical cases". There are two that may be used or adapted to GenAI from Moorkens (2022) in the open access book *Machine Translation for Everyone*. Case studies may be accompanied with a general question to motivate discussion (e.g. 'What ethical issues do you see in this scenario?') or a case worksheet featuring a number of questions or tasks. Some examples that could be included are:

- Describe the key ethical issues and the stakeholders involved.
- What ethical theories might provide guidance in this scenario?
- Can you think of guidelines from a translator code of ethics that could be useful in this scenario?
- Are there any legal issues involved in this scenario?
- With the above questions in mind, are there any actions that you propose to resolve this scenario in an ethical manner?

Students may address these questions in groups or as part of a plenary discussion moderated by the instructor. A final recap, making note of any new or unexpected issues that were brought up by students, can effectively summarise a class and reinforce key lessons.

5.3 Pedagogy of the implicated

The approach proposed by Bryan (2022) in her article on 'pedagogy of the implicated' could work in combination with or instead of case studies. As mentioned, Bryan draws on Rothberg (2019) for her description of (most of) us as "implicated subjects", who can be "beneficiaries and *perpetuators* of systems that are not of [our] own making or that [we] have no direct ancestral attachment to" (Bryan 2022: 338, italics in original). A preamble explaining this complex entanglement of implication is a valuable precursor to discussion in the classroom. The prompt for this discussion could come from carefully chosen images or a mapping exercise to visualise these entangled relationships.

Bryan's (2022: 341) article is focused on climate change education, for which there is a tendency towards what she calls an "apocalyptic sublime aesthetic". This aesthetic, she argues, "positions viewers as mere voyeurs or passive spectators – rather than active agents or implicated subjects – in the unfolding chaos". Images of AI are perhaps generally less apocalyptic, but are similarly uninvolving. These science-fiction-inspired images will be familiar to those who have seen many visualisations of AI featuring stylised robots with fingers to their chins among gigantic circuit boards, concealing any human involvement, ethical dilemmas, or even any question about the smooth transition towards wise and sentient intelligent robots.

Dihal & Duarte (2023) summarise how typical stock images of AI are misleading for non-experts, in that they hide societal and environmental impacts, promote unrealistic expectations of AI capabilities, hide human accountability, and often promote stereotypical assumptions about gender, ethnicity and religion. This led them to start the website Better Images of AI[1] featuring artist-created images, collages and illustrations that foreground the issues hidden in regular stock images. These images are highly considered and thus make an excellent starting point for classroom discussion and reflection.

Bryan proposes a mapping exercise as part of a discussion of self-implication in a complex interconnected consumer-focused system, such as that behind GenAI, to prompt critical self-reflection. The main example that she uses is what she calls the Social Ecology of Responsibility Framework (SERF), inspired by Bronfenbrenner's (2009) theory of the ecology of human development. Bronfenbrenner was "concerned with the mechanisms, processes and conditions that shape individuals' development and devised a model that theorises the reciprocal interactions that occur within and between different nested environments (or systems) which in turn affect developmental outcomes" (Bryan 2022: 338).

The SERF can be a useful tool to visualise the interconnections between the individual actions of the 'implicated subject' within their immediate context and larger social, national and international systems. These are represented by concentric circles, expanding from the individual's microsystem (our immediate context) to our macrosystem of societal, political, and cultural norms and context. Within these micro- and macrosystems are the mesosystem, describing points of connection between different levels, and the exosystem of institutions and organisations that exert influence without any direct contact with us as individuals.

[1] https://betterimagesofai.org

Navarro & Tudge (2023) suggest that our microsystems should be subdivided into physical and virtual categories, as our day-to-day activities, social roles, and interpersonal relations may differ (or blur) between physical and digitally-mediated domains. The mesosystem groups our multiple microsystems together. We might consider the remarkably consistent messaging about technology and GenAI coming from the exo- and macrosystems, why that might be, and the potential impacts on our mesosystem.

Figure 1 features an adaptation of Bryan's (2022) SERF, using a layered funnel design and bringing in the idea of the triple bottom line. This could form a useful starting point for classroom discussions or an example for a mapping exercise. The intention is not to blame individuals – the chronosystem part of the SERF makes clear the historical context for current systems – but to empower the individuals at the centre, which of course include the educators as elements and perpetuators within the overall system.

Figure 1: An adapted version of Bryan's (2022) Social Ecology of Responsibility Framework with some ideas to use for classroom discussion

Another useful image intended to stimulate reflection and discussion of the translator's responsibilities, with the potentially conflicting pressures from ethical codes, business ethics, personal interests, and social responsibility comes from Joseph Lambert's (2023) book *Translation Ethics*. The section on responsibilities and related questions place many of the points from this chapter within the context of translation work.

6 Conclusion

Many curricula now incorporate AI literacy as an important competence. Chan & Colloton (2024) recommend a fine-grained interpretation of AI literacy that is tailored to subject or professional areas, and a major part of this for translation students is AI ethics in translation. In this chapter, we briefly introduce some of the issues and debates regarding AI ethics, along with some positive initiatives that are ongoing.

We propose interactive, social constructivist methods for teaching students about AI ethics for translation, highlighting two potential methods in particular. The first of these is the use of carefully chosen case studies with accompanying support material to help students talk through ethical issues. Benbunan-Fich (1998: 21) feels that it is "crucial for students to practice ethical decision making, so when they have ethical decisions to make in the real business world they have a framework to follow". The second suggested method is an adaptation of Bryan's (2022) pedagogy of the implicated, using images and mindmapping to visualise and understand our own place within the tangle of beneficiaries and perpetrators of the system within which GenAI is developed and marketed. A disingenuous tendency to highlight individual responsibility for issues of ethics and sustainability lets large organisations off the hook, but these methods show how individual and collective responsibilities are interlinked. Our status as implicated subjects means that we must also take responsibility for our actions and ethical decisions. We hope that these methods will be effective beyond the individual class, leading to deeper knowledge in students that can come to their own individual ethical position on the debates and issues in AI ethics for translation.

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